

Uses of energy in living organisms:

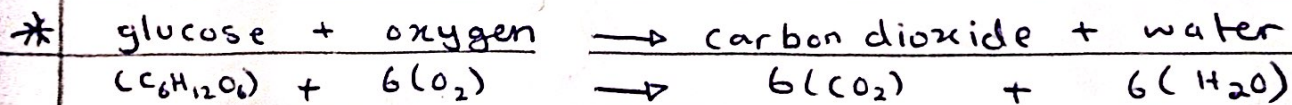
* muscle contraction, protein synthesis, cell division, active transport, growth, the passage of nerve impulses & the maintenance of a constant body temperature.

→ in the respiration of yeast, when temperature is increased, the rate of respiration increases as well because the heat speeds up the reactions (E_a is higher). Reaction speeds up and rate of cellular respiration increases. When temp. is decreased, cellular processes slow down.

AEROBIC RESPIRATION

aerobic respiration:

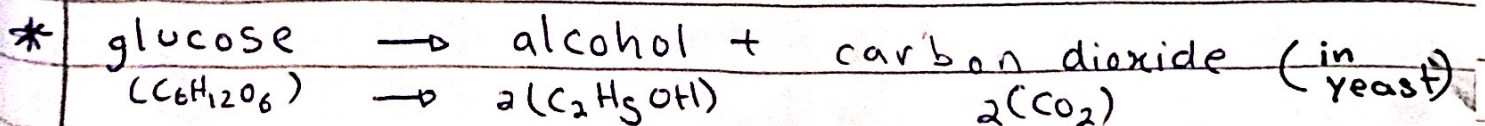
- the chemical reaction in cells that use oxygen to break down nutrients to release energy.



ANAEROBIC RESPIRATION

anaerobic respiration:

- the chemical reaction in cells that break down nutrients to release energy w/o oxygen.
- releases much less energy per glucose molecule than aerobic respiration



- glucose → Lactic acid (in muscles during vigorous exercise)
- lactic acid builds up in muscles & blood during vigorous exercise causing an oxygen debt.

How is the oxygen 'debt' removed:

- ↳ continuation of fast heart rate to transport lactic acid in the blood, from the muscles to the liver
- ↳ continuation of deeper/faster breathing to supply oxygen for aerobic respiration of lactic acid
- ↳ aerobic respiration of lactic acid in the liver